SMART INDIA HACKATHON 2026

SAGAR_SETU

Intelligent Freight Forecasting for East Coast Bulk Procurement

Problem Statement ID	SIH26006
Problem Statement	Development of an Intelligent Freight Forecasting Model for Optimized Vessel Chartering and Bulk Cargo Procurement from Overseas to East Coast of India
Theme	Transportation & Logistics
PS Category	Software
Team ID	**141436**
Team Name	**BLACK.OUT**

SAGAR_SETU: PROPOSED SOLUTION & APPROACH



FROM FRAGMENTED SIGNALS TO A CHARTER CASE

A workspace that explains the recommendation

01 **SEE**

Unify demand, vessel, freight, port and ocean context.

02 **TEST**

Compare laycan, vessel and cargo-allocation scenarios.

03 **DECIDE**

Send the strongest scenario to authorized analyst approval.

OUTPUT : An explainable charter plan with confidence, trade-offs and an approval trail.

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TECHNOLOGY STACK

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MVP stack aligned to the real SAGAR_SETU charter-decision workflow.

01. CORE STACK

FRONTEND — **React**
Decision workspace & analyst UI

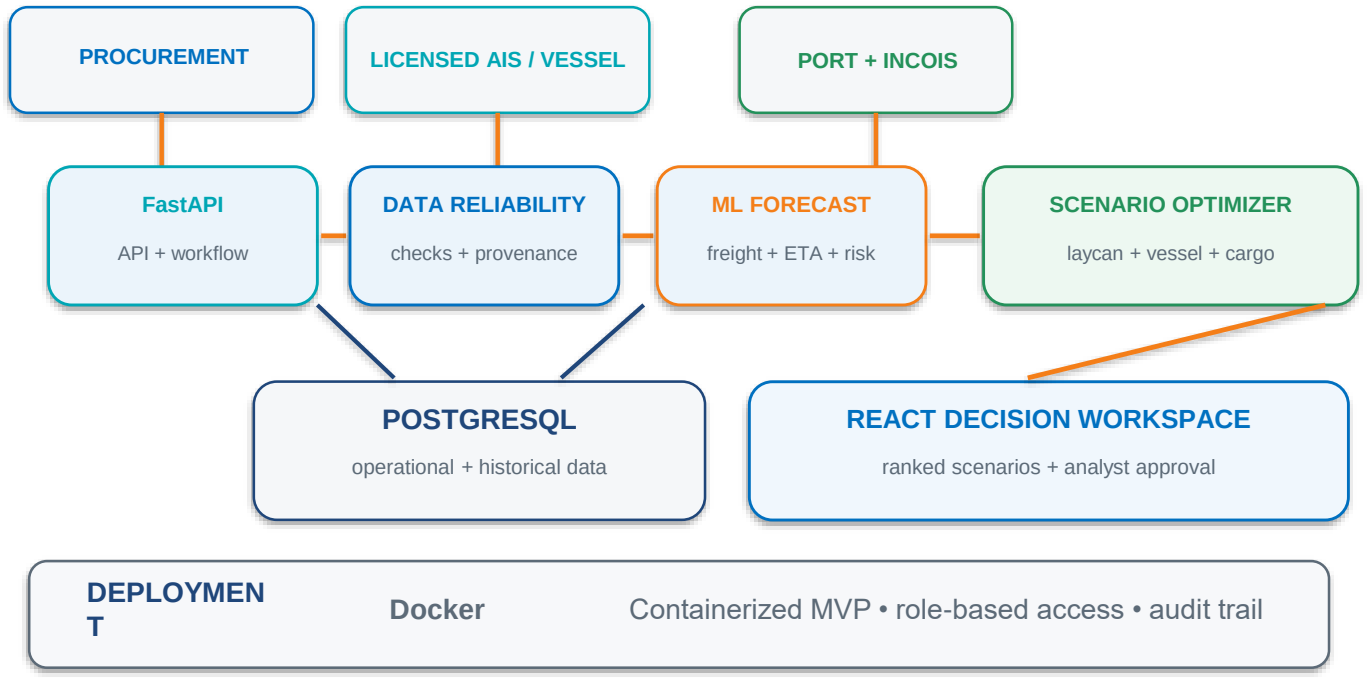
BACKEND / API — **FastAPI + Python**
Workflow, validation & service layer

DATA & ML — **ML**
Freight, ETA & risk forecasting

DATABASE — **PostgreSQL**
Operational & historical data

DEPLOYMENT — **Docker**
Containerized MVP deployment

02. SYSTEM FLOW



External data connections require authorization, licensing and data-sharing agreements.

FEASIBILITY AND IMPLEMENTATION

A staged pilot is practical because every critical risk has a designed response.

01 BUILDABLE

Use data already present in the workflow

- Procurement history + inventory cover
- Authorized port and ocean inputs
- Python + PostgreSQL MVP

02 VIABLE PILOT

Validate before expanding operations

- One cargo, one discharge port
- Historical backtest
- Shadow-mode recommendations

03 RISKS

Keep uncertainty visible

- Inconsistent operational records
- Rate and route disruptions
- Data access constraints

04 CONTROL PLAN

Keep the decision trustworthy

- Confidence ranges and quality gates
- Stress cases + human override
- Role access and audit log

IMPLEMENTATION PLAN:

PHASE 1 - SCOPE
Define cargo + port; map required data

→

PHASE 2 - BACKTEST
Train + validate model on historical voyages

→

PHASE 3 - SHADOW
Run parallel decisions; analyst reviews outputs

→

PHASE 4 - PILOT
Controlled rollout; monitor and refine

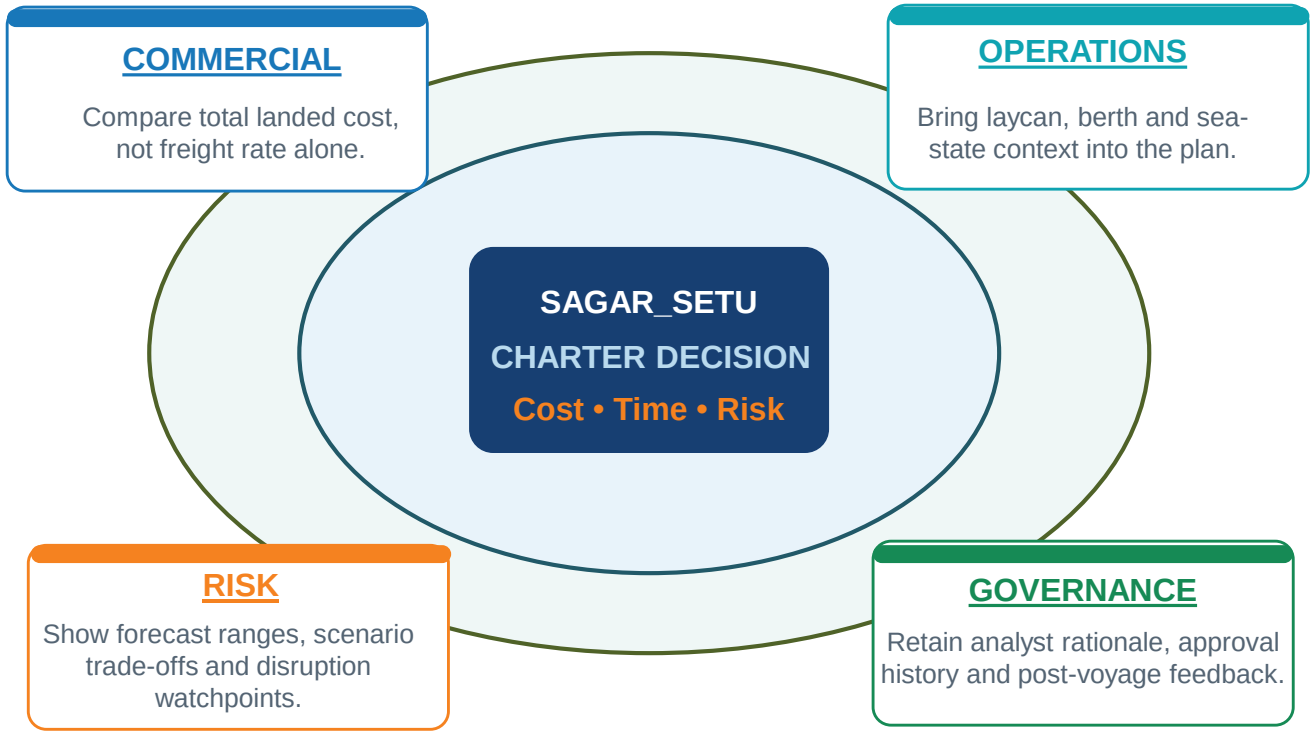
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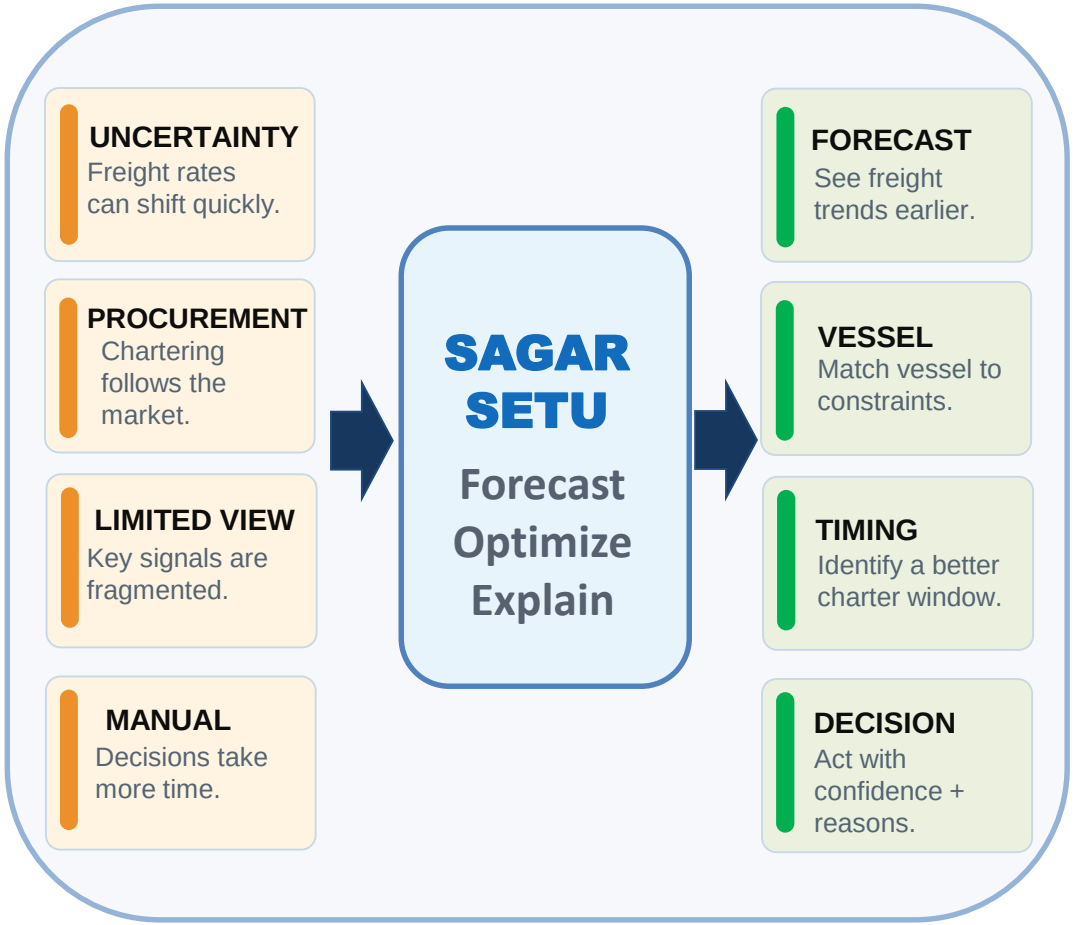
IMPACT AND BENEFITS

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The value is a stronger decision process, measured against the current manual baseline.

WHAT SAGAR_SETU IMPROVES





RESEARCH AND REFERENCES

Official sources used to frame the problem, operating context and feasible data integrations

01	**Ministry of Steel Annual Report 2024-25**	Coking coal import dependence and FY 2023-24 import volume.	https://steel.gov.in/node/2033
02	**Visakhapatnam Port Authority Report 2024-25**	East Coast pilot context and 82.62 MT cargo throughput.	https://shipmin.gov.in/en/content/visakhapatnam-port-authority-report-english
03	**MoPSW Information Technology**	Maritime Single Window - Sagarsetu and authorized ship-data exchange.	https://shipmin.gov.in/en/content/information-technology
04	**INCOIS Ocean State Forecast**	Forecast inputs for waves, winds, currents and sea-state risk.	https://incois.gov.in/site/services/osf.jsp
05	**UNCTAD Review of Maritime Transport 2025**	Freight-rate volatility, rerouting and dry-bulk market context.	https://unctad.org/publication/review-maritime-transport-2025

Note: Live data connections require data-sharing approval, API licensing and role-based access controls during implementation.